

- If patient is taking dextromethorphan (serotonin reuptake inhibitor)
- If patient is taking diuretics or other antihypertensives without careful monitoring of blood pressure
- If patient has pheochromocytoma
- If patient has cardiovascular or cerebrovascular disease, unless cleared by the patient's medical doctor
- If patient is taking a prohibited drug
- If patient is not compliant with a low-tyramine diet
- If there is a proven allergy to phenelzine

Special Populations

Renal Impairment

- Use with caution –drug may accumulate in plasma
- May require lower than usual adult dose

Hepatic Impairment

- Phenelzine should be used cautiously

Cardiac Impairment

- Contraindicated in patients with congestive heart failure or hypertension, unless cleared by the patient's medical doctor

Elderly

- Initial dose 7.5 mg/day; increase every few days by 7.5–15 mg/day
- Elderly patients may have greater sensitivity to adverse effects, but many tolerate MAOIs
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents

- Not recommended for use under age 16
- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester
- Possible increased incidence of fetal malformations if phenelzine is taken during the first trimester
- Should evaluate patient for treatment with an antidepressant with a better risk/benefit ratio
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Some drug is found in mother's breast milk
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Should evaluate patient for treatment with an antidepressant with a better risk/benefit ratio

The Art of Psychopharmacology

Potential Advantages

- Atypical depression
- Severe depression
- Treatment-resistant depression or anxiety disorders

Potential Disadvantages

- Requires compliance to dietary restrictions, concomitant drug restrictions
- Patients with cardiac problems or hypertension, unless cleared by the patient's medical doctor
- Multiple daily doses, although many patients can be dosed more than once a day

Primary Target Symptoms

- Depressed mood
- Somatic symptoms
- Sleep and eating disturbances
- Psychomotor retardation
- Morbid preoccupation
- Anxiety

Pearls

- Excellent anxiolytic effects often observed

- Generally more weight gain and sexual dysfunction than with tranylcypromine
 - MAOIs are generally reserved for third- or fourth-line use after SSRIs, SNRIs, and combinations of newer antidepressants have failed
 - Patient should be advised not to take any prescription or over-the-counter drugs without consulting their doctor because of possible drug interactions with the MAOI
 - Headache is often the first symptom of hypertensive crisis
 - The rigid dietary restrictions may reduce compliance (see [Table 2](#) after Pearls)
 - Mood disorders can be associated with eating disorders (especially in adolescent females), and phenelzine can be used to treat both depression and bulimia
- * Myths about the danger of dietary tyramine can be exaggerated, but prohibitions against concomitant drugs often not followed closely enough
- Orthostatic hypotension, insomnia, and sexual dysfunction are often the most troublesome common side effects
- * MAOIs should be for the expert, especially if combining with agents of potential risk (e.g., stimulants, trazodone, TCAs)
- * MAOIs should not be neglected as therapeutic agents for the treatment-resistant
- For treatment-resistant patients, experts can give a tricyclic/tetracyclic antidepressant other than clomipramine or imipramine simultaneously

with an MAOI for patients who fail to respond to numerous other antidepressants

- Use of MAOIs with clomipramine is always prohibited because of the risk of serotonin syndrome and death due to serotonin reuptake inhibition
- Amoxapine may be the preferred tricyclic/tetracyclic antidepressant to combine with an MAOI in heroic cases due to its theoretically protective 5HT_{2A} antagonist properties
- If this option is elected, start the MAOI with the tricyclic/tetracyclic antidepressant simultaneously at low doses after appropriate drug washout, then alternately increase doses of these agents every few days to a week as tolerated
- Although very strict dietary and concomitant drug restrictions must be observed to prevent hypertensive crises and serotonin syndrome, the most common side effects of MAOI and tricyclic/tetracyclic combinations may be weight gain and orthostatic hypotension

Suggested Reading

Gillman K . PsychoTropical Research (PTR). <https://psychotropical.com> .

Gillman PK . A reassessment of the safety profile of monoamine oxidase inhibitors: elucidating tired old tyramine myths . *J Neural Transm (Vienna)* 2018 ;**125** (11):1707–17 .

Kennedy SH . Continuation and maintenance treatments in major depression:
the neglected role of monoamine oxidase inhibitors . *J Psychiatry Neurosci*
1997 ;**22** :127–31 .

Phentermine/topiramate

Therapeutics

Brands

- Qsymia
- *see index for additional brand names*

Generic?

No

Class

- Norepinephrine and dopamine reuptake inhibitor (phentermine) combined with a voltage-sensitive sodium channel modulator (topiramate); weight management medication

Commonly Prescribed for

(bold for FDA approved)

- **Chronic weight management (adjunct to reduced-calorie diet and increased physical activity) in adults with an initial BMI of at least 30 kg/m² (obese) or at least 27 kg/m² (overweight) in the presence of at least 1 weight-related comorbid condition**

How the Drug Works

- Phentermine increases dopamine and norepinephrine by blocking both the dopamine and the norepinephrine transporters. In the hypothalamus, these 2 neurotransmitters activate POMC neurons, causing the release of POMC. POMC is then broken down into α -melanocyte-stimulating hormone, which binds to melanocortin 4 receptors to suppress appetite. However, stimulation of POMC neurons also activates an endogenous opioid-mediated negative feedback loop, which mitigates the appetite-suppressing effects
- Theoretically, through modulation of voltage-sensitive sodium channels topiramate may reduce glutamatergic stimulation and increase GABAergic inhibition in the appetite-stimulating pathway, resulting in net inhibition of this pathway. Such an action would synergize with simultaneous activation of the appetite-suppressing pathway by phentermine. This results in more robust and long-lasting appetite suppression than with either drug alone

How Long Until It Works

- At least 3% weight loss is generally achieved after 14 weeks; at least 5% weight loss is generally achieved after an additional 12 weeks on maximum daily dose

If It Works

- Patients may achieve 5–10% reduction from baseline in body weight

If It Doesn't Work

- Discontinue if 5% weight loss is not achieved after 12 weeks on maximum daily dose

Best Augmenting Combos for Partial Response or Treatment Resistance

- Phentermine/topiramate itself should be administered in conjunction with reduced-calorie diet and increased physical activity
- Often best to try another strategy prior to resorting to augmentation

Tests

- Women of childbearing potential should have a negative pregnancy test before starting phentermine/topiramate and monthly during treatment
- Regular measurement of resting heart rate, especially for patients with cardiac or cerebrovascular disease or when initiating or increasing the dose of phentermine/topiramate
- Obtain a blood chemistry profile that includes bicarbonate, creatinine, potassium, and glucose at baseline and periodically during treatment
- Measurement of blood pressure prior to starting phentermine/topiramate and during phentermine/topiramate treatment is recommended in patients being treated for hypertension

Side Effects

How Drug Causes Side Effects

- CNS side effects theoretically due to excessive actions at voltage-sensitive sodium channels or excessive actions on norepinephrine and dopamine
- Weak inhibition of carbonic anhydrase may lead to kidney stones and paresthesias
- Inhibition of carbonic anhydrase may also lead to metabolic acidosis

Notable Side Effects

- Constipation, dry mouth
- Paresthesia, dizziness, dysgeusia, insomnia
- Cognitive impairment (may be more likely with rapid titration or high initial doses)



Life-Threatening or Dangerous Side Effects

- Hypoglycemia
- Increased heart rate
- Metabolic acidosis
- Kidney stones
- Secondary angle-closure glaucoma
- Oligohidrosis and hyperthermia (more common in children)
- Sudden unexplained deaths have occurred in epilepsy (unknown if related to topiramate use)

- Rare activation of suicidal ideation and behavior (suicidality)

Weight Gain



- Reported but not expected

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait
- Avoid dosing in evening due to the possibility of insomnia
- In a few weeks, switch to another agent

Best Augmenting Agents for Side Effects

- Often best to try another treatment prior to resorting to augmentation strategies to treat side effects

Dosing and Use

Usual Dosage Range

- 7.5 mg/46 mg per day up to 15 mg/92 mg per day

Dosage Forms

- Capsule (phentermine mg/topiramate mg extended-release) 3.75 mg/23 mg, 7.5 mg/46 mg, 11.25 mg/69 mg, 15 mg/92 mg

How to Dose

- Initial 3.75 mg/23 mg per day; after 14 days increase to 7.5 mg/46 mg per day
- If at least 3% weight loss is not achieved after 12 weeks on 7.5 mg/46 mg per day, can increase to 11.25 mg/69 mg per day; after 14 days increase to 15 mg/92 mg per day
- Discontinue if at least 5% weight loss is not achieved after 12 weeks on 15 mg/92 mg per day



Dosing Tips

- Can be taken with or without food
- Side effects tend to abate with time but may recur again briefly with dose increases

Overdose

- Phentermine: restlessness, tremor, hyperreflexia, rapid respiration, confusion, aggression, hallucinations, panic
- Topiramate: severe metabolic acidosis, convulsions, sedation, speech disturbance, blurred or double vision, impaired coordination, hypotension, abdominal pain, agitation, dizziness

Long-Term Use

- Has been evaluated in controlled studies up to 1 year

Habit Forming

- Phentermine is a Schedule IV drug
- Topiramate is not controlled in the Controlled Substances Act

How to Stop

- For patients taking 15 mg/92 mg per day, take a dose every other day for at least 1 week prior to discontinuing treatment, due to the possibility of precipitating a seizure

Pharmacokinetics

- Phentermine mean terminal half-life approximately 20 hours
- Topiramate mean terminal half-life approximately 65 hours
- Neither phentermine nor topiramate is extensively metabolized



Drug Interactions

- May cause irregular bleeding in patients taking oral contraceptives, but does not increase risk of pregnancy; patients should be advised not to discontinue oral contraceptives if spotting occurs
- May increase effects of CNS depressants

- May potentiate hypokalemia in patients taking non-potassium-sparing diuretic agents
- Weight loss associated with phentermine/topiramate may increase the risk of hypotension in patients taking antihypertensive medications
- Carbamazepine, phenytoin, and valproate may increase the clearance of topiramate, and thus decrease topiramate levels, possibly requiring a higher dose of topiramate
- Topiramate may increase the clearance of phenytoin and thus decrease phenytoin levels, possibly requiring a higher dose of phenytoin
- Topiramate may increase the clearance of valproate and thus decrease valproate levels, possibly requiring a higher dose of valproate
- Topiramate may increase plasma levels of metformin; also, metformin may reduce clearance of topiramate and increase topiramate levels
- Topiramate may interact with carbonic anhydrase inhibitors to increase the risk of kidney stones
- Reports of hyperammonemia with or without encephalopathy in patients taking topiramate combined with valproate, though this is not due to a pharmacokinetic interaction; in patients who develop unexplained lethargy, vomiting, or change in mental status, an ammonia level should be measured

- Phentermine is contraindicated with MAOIs due to risk of hypertensive crisis

Other Warnings/Precautions

- Phentermine/topiramate can cause fetal toxicity; women of childbearing potential should obtain a negative pregnancy test before treatment and monthly thereafter and use effective contraception
- If a patient develops a sustained increase in resting heart rate while taking phentermine/topiramate, the dose should be reduced or the medication should be discontinued
- May cause an increase in serum creatinine; if persistent elevations in creatinine occur during phentermine/topiramate treatment, the dose should be reduced or the medication should be discontinued
- Phentermine/topiramate has not been studied in combination with insulin; weight loss may increase the risk of hypoglycemia in patients with type 2 diabetes mellitus treated with insulin and/or insulin secretagogues, and medication adjustments may be necessary
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such side effects immediately

Do Not Use

- If patient is pregnant

- If patient has glaucoma
- If patient has hyperthyroidism
- If patient is taking an MAOI
- If there is a proven allergy to phentermine or topiramate

Special Populations

Renal Impairment

- Maximum dose is 7.5 mg/46 mg per day in patients with moderate to severe impairment
- Not studied or recommended for use in patients with end-stage renal disease

Hepatic Impairment

- Maximum dose is 7.5 mg/46 mg per day in patients with moderate impairment
- Not recommended for use in patients with severe impairment

Cardiac Impairment

- Not systematically evaluated in patients with cardiac impairment
- Not recommended for patients with recent or unstable cardiac or cerebrovascular disease

Elderly

- Some patients may tolerate lower doses better



Children and Adolescents

- Safety and efficacy have not been established
- Not recommended for use in children or adolescents



Pregnancy

- Contraindicated
- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- There is an increased risk of cleft lip/palate with topiramate

Breast Feeding

- Some drug is found in mother's breast milk
- Recommended to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Patients who have had past weight loss on either active compound alone may experience a more robust weight loss effect and better tolerability on the combination of phentermine with topiramate than with administration of either drug alone

Potential Disadvantages

- Patients with a history of kidney stones or risks for metabolic acidosis
- Women who are pregnant or wish to become pregnant
- Patients with a history of past stimulant abuse

Primary Target Symptoms

- Excess weight

Pearls

- Has perhaps the largest effect size on weight loss of drugs available for the treatment of obesity
- Could theoretically be effective in binge eating disorder
- Because of the teratogenic risk with phentermine/topiramate, it is available only through a Risk Management and Mitigation Strategy (REMS) program; only certified pharmacies may distribute it

Suggested Reading

Colman E , Golden J , Roberts M , et al. The FDA's assessment of two drugs for chronic weight management . *N Engl J Med* 2012 ;**367** (17):1577–9 .

.....

Cosentino G , Conrad AO , Uwaifo GI . Phentermine and topiramate for the management of obesity: a review . *Drug Des Devel Ther* 2011 ;**7** :267–87 .

.....

Holes-Lewis KA , Malcolm R , O'Neil PM . Pharmacotherapy of obesity: clinical treatments and considerations . *Am J Med Sci* 2013 ;**345** (4):284–8 .

.....

Shin JH , Gadde KM . Clinical utility of phentermine/topiramate (QsymiaTM) combination for the treatment of obesity . *Diabetes Metab Syndr Obes* 2013 ;**6** :1319 .

Pimavanserin

Therapeutics

Brands

- Nuplazid

see index for additional brand names

Generic?

No

Class

- Atypical antipsychotic; serotonin 2A/2C antagonist/inverse agonist

Commonly Prescribed for

(bold for FDA approved)

- **Hallucinations and delusions associated with Parkinson's disease psychosis**
- Dementia-related psychosis

How the Drug Works

- * Antagonism/inverse agonism at 5HT_{2A} receptors

- Pimavanserin is also an antagonist/inverse agonist at 5HT2C receptors (activity is very low compared to that at 5HT2A receptors)

How Long Until It Works

- In a clinical trial, psychotic symptom improvement reached significance within 1 month

If It Works

- Most often reduces hallucinations and delusions without worsening parkinsonism

If It Doesn't Work

- Try reducing the dose of antiparkinsonian dopaminergic therapies
- Try low-dose quetiapine
- Try low-dose clozapine
- Do not try any other atypical or conventional antipsychotic



Best Augmenting Combos for Partial Response or Treatment Resistance

- None known
- Theoretically, might be possible to combine with low-dose quetiapine, especially during prolonged cross-titration when switching from one to the other

Tests

- None

Side Effects

How Drug Causes Side Effects

- Mechanism of peripheral edema, confusional state, and nausea unknown

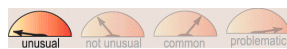
Notable Side Effects

- Peripheral edema
- Confusional state
- Nausea



Life-Threatening or Dangerous Side Effects

- QTc prolongation
- Increased risk of death and cerebrovascular events in elderly patients with dementia-related psychosis has occurred with antipsychotic use



Weight Gain

- Reported but not expected



Sedation

- Reported but not expected

What to Do About Side Effects

- Wait
- Wait
- Wait
- Discontinue if side effects are intolerable

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 34 mg once daily

Dosage Forms

- Tablet 10 mg
- Capsule 34 mg

How to Dose

- 34 mg/day taken once daily; no titration required



Dosing Tips

- Can be taken with or without food
- No dose adjustment of carbidopa/levodopa is required

Overdose

- Limited experience

Long-Term Use

- Not studied, but long-term maintenance treatment is often necessary for Parkinson's disease psychosis

Habit Forming

- No

How to Stop

- The long half-life suggests that it may be possible to stop pimavanserin abruptly
- Rapid discontinuation could theoretically lead to rebound psychosis and worsening of symptoms, but less likely with pimavanserin due to its long half-life

Pharmacokinetics

- Mean half-life 57 hours (pimavanserin) and 200 hours (active metabolite N-desmethylated metabolite)